

XIN BARAG YOUQI, China

WMO: 506030

Lat: 48.6782N

Lon: 116.8100E

Elev: 557

StdP: 94.81

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.5	-29.3	-35.2	0.1	-31.1	-33.1	0.2	-28.8	11.7	-15.5	10.1	-16.6	2.3	250	0.618

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	33.1	18.1	31.2	17.5	29.2	17.0	20.6	27.5	19.8	26.4	19.1	25.4	4.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.7	14.4	22.8	17.8	13.6	22.6	16.8	12.8	22.1	62.4	27.3	59.4	26.4	56.9	25.4	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.4	9.5	8.1	-33.0	36.9	3.2	2.7	-35.3	38.9	-37.2	40.5	-39.0	42.0	-41.3	44.0		
(5)			-33.1	22.1	3.2	0.9	-35.4	22.7	-37.2	23.2	-39.0	23.7	-41.3	24.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.1	-21.0	-16.0	-6.8	4.4	12.7	19.7	22.5	20.1	12.9	2.9	-9.4	-18.4		
	DBStd	15.96	4.71	5.30	6.26	4.99	4.59	4.04	2.99	3.37	4.52	5.07	6.32	4.85		
	HDD10.0	4120	961	728	519	178	24	0	0	0	21	225	583	882		
	HDD18.3	6234	1219	961	778	417	184	31	3	19	171	479	833	1140		
	CDD10.0	1221	0	0	0	11	107	291	386	313	108	4	0	0		
	CDD18.3	292	0	0	0	0	8	71	131	74	8	0	0	0		
	CDH23.3	2995	0	0	0	11	167	808	1228	669	112	1	0	0		
	CDH26.7	1096	0	0	0	2	52	317	468	226	31	0	0	0		
(14) Wind	WSAvg	3.5	3.1	3.3	3.8	4.4	4.4	3.4	3.1	3.2	3.5	3.5	3.2	3.0		
(15) Precipitation	PrecAvg	250	1	1	2	6	12	39	80	71	28	5	2	1		
	PrecMax	446	3	9	8	21	42	87	187	173	93	25	12	4		
	PrecMin	103	0	0	0	0	0	3	13	18	0	0	0	0		
	PrecStd	76	1	2	2	6	11	24	39	45	22	6	3	1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.8	1.7	13.7	24.8	31.5	36.6	36.2	34.4	30.5	21.3	8.4	-3.1		
		MCWB	-7.7	-2.8	4.8	10.3	14.0	17.2	19.5	18.5	15.5	10.8	2.6	-5.2		
	2%	DB	-9.4	-2.4	9.5	20.0	27.8	32.8	33.2	31.4	26.7	16.5	4.8	-6.8		
		MCWB	-10.7	-5.2	2.3	7.8	12.6	16.8	19.2	17.8	14.3	7.7	0.3	-8.3		
	5%	DB	-11.9	-5.3	5.8	16.6	24.6	30.4	31.2	29.1	23.7	13.5	2.0	-9.3		
		MCWB	-13.0	-7.3	0.0	6.3	11.6	16.4	18.8	17.5	13.4	6.1	-1.5	-10.5		
	10%	DB	-14.3	-7.9	3.0	13.6	21.9	27.9	29.2	27.0	21.2	11.0	-0.6	-11.5		
		MCWB	-15.1	-9.4	-1.5	4.8	10.3	15.6	18.2	17.2	12.5	4.9	-3.2	-12.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-7.7	-2.5	5.0	10.6	15.5	19.8	22.2	21.0	18.1	10.7	2.8	-5.1		
		MCDB	-6.0	1.4	13.5	24.0	27.3	28.1	29.3	27.9	23.4	20.6	8.1	-3.3		
	2%	WB	-10.6	-5.1	2.4	8.3	13.7	18.6	21.2	20.0	16.2	8.3	0.5	-8.2		
		MCDB	-9.4	-2.5	9.1	19.1	24.7	26.5	28.1	26.0	22.2	14.8	4.6	-6.8		
	5%	WB	-12.9	-7.3	0.3	6.5	12.4	17.7	20.3	19.3	14.8	6.7	-1.3	-10.4		
		MCDB	-11.9	-5.4	5.4	15.8	22.5	26.4	27.2	25.3	21.7	12.6	1.7	-9.3		
	10%	WB	-15.1	-9.4	-1.3	5.0	11.2	16.9	19.6	18.5	13.3	5.3	-3.1	-12.4		
		MCDB	-14.3	-8.0	2.8	13.1	20.5	25.6	26.3	24.3	20.1	10.6	-0.7	-11.6		
(35) Mean Daily Temperature Range	MDBR		9.0	11.0	11.8	12.7	13.0	11.7	10.6	10.9	12.0	11.1	9.8	8.5		
	5% DB	MCDBR	11.4	13.6	15.4	17.9	17.9	15.6	14.2	14.2	15.8	15.2	12.3	10.7		
		MCWBR	10.3	11.2	9.9	9.3	7.8	5.5	4.6	4.9	7.0	8.7	8.6	9.5		
	5% WB	MCDBR	11.4	13.5	14.7	16.9	15.5	12.0	10.9	10.5	12.6	13.8	12.0	10.7		
		MCWBR	10.3	11.1	9.6	9.0	7.1	4.5	4.2	4.4	6.1	8.2	8.5	9.4		
(40) Clear-Sky Solar Irradiance	taub		0.260	0.283	0.327	0.403	0.485	0.472	0.434	0.394	0.346	0.319	0.285	0.267		
	taud		2.250	2.211	2.179	2.068	1.880	2.042	2.226	2.306	2.374	2.353	2.274	2.205		
	Ebn at Noon		806	867	879	840	781	796	824	843	855	817	761	732		
	Edh at Noon		73	98	121	151	190	163	134	118	99	85	72	68		
(44) All-Sky Solar Radiation	RadAvg		1.85	3.10	4.50	5.61	6.41	6.58	6.14	5.59	4.50	3.13	1.93	1.43		
	RadStd		0.11	0.17	0.26	0.28	0.32	0.45	0.36	0.37	0.27	0.16	0.10	0.11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	-2.27	-3.01	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		N/A	N/A	-2.65	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page